

Agentic RAG Chatbot — Project Overview and Setup Guide

The Agentic RAG Chatbot is a powerful AI-based tool that allows users to upload various documents like PDFs, Word files, CSVs, text files, and PowerPoints and ask questions about them. The chatbot reads and understands the content of these files, retrieves the most relevant parts, and then uses OpenAI's GPT-4 to generate clear, context-aware answers. What makes this chatbot special is that it's built using an agent-based architecture, where each agent has a dedicated role: one for ingestion (reading files), one for retrieval (finding relevant information), and one for response generation (answering the user's question). It also supports Model Context Protocol (MCP) to make communication between these agents structured and modular.

What Is This Thing?

Imagine having a personal assistant that can read through all your documents—PDFs, Word files, spreadsheets, presentations and then answer any questions you have about them. That's exactly what this Agentic RAG Chatbot does! we simply upload our files, ask questions in plain English, and the chatbot gives you accurate answers based on what it found in your documents. It even tells you which specific files it used to answer your question, so you know the information is reliable.

What Makes It Special?

Think of this chatbot as having three specialized team members working together:

- **The Reader** - Takes your uploaded files and carefully reads through everything, breaking content into digestible pieces
- **The Searcher** - When you ask a question, this agent quickly finds the most relevant information from all your documents
- **The Answerer** - Takes the relevant information and uses advanced AI (GPT-4) to craft a clear, helpful response

This team approach makes the system really smart and easy to improve over time.

Getting Started

Step 1: Get the Project Files

If you're comfortable with Git, clone the repository:

```
bash
```

```
git clone https://github.com/your-username/agentic-rag-chatbot.git
```

```
cd agentic-rag-chatbot
```

If Git feels intimidating, just download the project as a ZIP file and extract it to a folder on your computer.

Step 2: Set Up Your Workspace (Optional but Smart)

Think of this like creating a clean workspace just for this project:

```
bash
```

```
python -m venv venv
```

Then activate your workspace:

- **Windows users:** `venv\Scripts\activate`
- **Mac/Linux users:** `source venv/bin/activate`

Step 3: Install the Tools You Need

This is like gathering all the tools before starting a DIY project:

```
bash
```

```
pip install -r requirements.txt
```

If that doesn't work, try installing the essentials manually:

```
bash
```

```
pip install streamlit sentence-transformers openai numpy
```

Step 4: Get Your AI Key

The chatbot uses OpenAI's GPT-4 to generate smart answers. You'll need an API key from OpenAI (think of it like a subscription):

Easy way: Set it in your system

```
bash
```

```
export OPENAI_API_KEY=your-actual-api-key-here
```

Quick way: Edit the response_agent.py file and replace "your-api-key" with your real key (just remember this isn't secure for sharing with others)

Step 5: Launch Your Chatbot!

Ready for the magic moment? Start your chatbot:

```
bash
```

```
streamlit run app.py
```

Your web browser should automatically open with a friendly interface where you can start uploading documents and asking questions!

How to Use It

1. **Upload Your Documents** - Drag and drop or browse for PDFs, Word docs, text files, CSVs, or PowerPoint presentations
2. **Ask Questions** - Type any question about your documents in natural language
3. **Get Smart Answers** - The chatbot will analyze your documents and provide detailed answers
4. **See the Sources** - You'll know exactly which documents were used to answer your question

Behind the Scenes Magic

The system works like a well-organized research team:

1. **Document Processing:** When you upload files, the Ingestion Agent carefully reads everything and organizes the content into searchable chunks
2. **Smart Search:** When you ask a question, the Retrieval Agent uses advanced semantic search to find the most relevant information (it understands meaning, not just keywords!)
3. **Intelligent Responses:** The Response Agent takes the relevant information and uses GPT-4 to craft a comprehensive answer in natural language
4. **Structured Communication:** All these agents talk to each other using a standardized protocol, making everything run smoothly.